

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [150, 152, 146, 163, 163, 161, 154, 167, 156, 171]

Tap amplitudes (pixels): [127.94139283281233, 122.80065146407001, 126.21014222319853, 94.5568612000208, 111.07204868912791, 117.6860229593982, 95.1892851112981, 88.3911760301898, 94.57801012920498, 102.45974819410792, 78.39005038906404]

GPT-5

Analysis of extracted features for MDS-UPDRS Item 3.4 (Finger Tapping)

- Instruction target: 10 taps. Extracted shows Number of taps = 11, with 10 inter-tap periods provided (typical if an extra peak was detected). I'll evaluate the first 10 taps, while noting any over-tapping as a potential sign of rhythm irregularity.

Speed (periods)

- Tap periods (frames): [150, 152, 146, 163, 163, 161, 154, 167, 156, 171]
- These are relatively consistent with a mild trend toward slowing over the sequence (last period 171 vs early ~146–152). No marked bradykinesia (moderate) is evident; rather, slight-to-mild slowing with small increases in period.

Rhythm/Interruptions

- The sequence of periods is smooth without large outliers or pauses. No evidence of >5 interruptions or a clear long arrest. There may be 0–2 minor hesitations suggested by modest variability, but not enough to qualify for “3–5 interruptions.”

Amplitude and decrement

- Tap amplitudes (pixels), 11 values: [127.9, 122.8, 126.2, 94.6, 111.1, 117.7, 95.2, 88.4, 94.6, 102.5, 78.4]
- Pattern: High initial amplitudes (~123–128), then a notable drop by tap 4 (~94.6), partial recovery (111–118), then recurring lower amplitudes (95, 88, 95), brief recovery (~102), and the final tap is the lowest (78.4).
- Decrement clearly begins well before the end—after the first few taps (by tap 4), with fluctuating but generally reduced amplitudes thereafter. This aligns with “amplitude decrements midway through the 10-tap sequence” (score 2). It is not “starting after the 1st tap” consistently enough to claim score 3, given partial recoveries and not a monotonic early decline from tap 2 onward.

Overall mapping to criteria

- Interruptions: none to few; not in the 3–5 range.
- Speed: slight to mild slowing, not moderate.
- Amplitude: decrement emerges by about the mid portion of the sequence and persists with variability, matching the “midway” criterion.

Score

2
